Revision Notes

CBSE Class 12 Chemistry • NCERT Part I, Chapters 1–5

Solutions • Electrochemistry • Chemical Kinetics • d- and f-Block Elements • Coordination Compounds

How to use these notes. Keep the separate **formula sheet** next to you. Each chapter has: mechanisms / derivations or key reactions; a **give-reasons bank** or conceptual questions; **exercise-wise solution patterns** (every NCERT exercise grouped into patterns, each with one solved question from the book, and a small line listing which questions belong to which pattern); exam tips; purple **JEE extras** (not needed for the board exam); and a **pattern spotter** for final revision. Question numbers match the NCERT textbook (Reprint 2026–27).

Contents

Chapter 1 Solutions **2**

1A Derivations (board favourites) . . . 2
1B Diagrams and graphs to practise . . . 3
1C Exercise-wise numerical patterns . . . 3
1D Conceptual / reasoning questions . . . 4
1E Exam tips . . . 4
1F JEE extras . . . 4
1G Pattern spotter . . . 5

Chapter 2 Electrochemistry **6**

2A Derivations (board favourites) . . . 6
2B Diagrams and graphs to practise . . . 6
2C Exercise-wise numerical patterns . . . 7
2D Conceptual / reasoning questions . . . 8
2E Exam tips . . . 8
2F JEE extras . . . 8
2G Pattern spotter . . . 8

Chapter 3 Chemical Kinetics **10**

3A Derivations (board favourites) . . . 10
3B Diagrams and graphs to practise . . . 10
3C Exercise-wise numerical patterns . . . 11
3D Conceptual / reasoning questions . . . 12
3E Exam tips . . . 12
3F JEE extras . . . 12
3G Pattern spotter . . . 12

Chapter 4 The d- and f-Block Elements **14**

4A Give-reasons bank 14
4B Exercise-wise solution patterns 16
4C Exam tips 17
4D JEE extras 18
4E Pattern spotter 19

Chapter 5 Coordination Compounds 20
5A Give-reasons bank 20
5B Exercise-wise solution patterns 21
5C Exam tips 22
5D JEE extras 22
5E Pattern spotter 22

Chapter 1 Solutions

1A Derivations (board favourites)

1. Relative lowering of vapour pressure ⇒ molar mass

Raoult: $p_1 = x_1 p_1^\circ$ (solvent), non-volatile solute. $\dfrac{p_1^\circ - p_1}{p_1^\circ} = x_2 = \dfrac{n_2}{n_1 + n_2} \approx \dfrac{n_2}{n_1}$ (dilute)

$$\Rightarrow \boxed{\frac{p_1^\circ - p_1}{p_1^\circ} = \frac{w_2 M_1}{M_2 w_1}}.$$

2. Elevation of boiling point / depression of freezing point

Both are proportional to molality: $\Delta T_b = K_b m$, $\Delta T_f = K_f m$, $m = \dfrac{w_2 \times 1000}{M_2 \times w_1}$

$$\Rightarrow \boxed{M_2 = \frac{K_f \times w_2 \times 1000}{\Delta T_f \times w_1}}$$ (same with K_b, ΔT_b).

3. Osmotic pressure

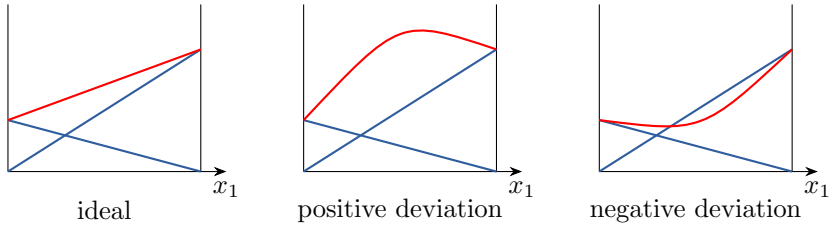
$\pi = CRT = \dfrac{n_2}{V} RT \Rightarrow \boxed{M_2 = \dfrac{w_2 RT}{\pi V}}$. Preferred for macromolecules (measurable at room temperature, large value even for dilute solutions).

4. van't Hoff factor

$i = \dfrac{\text{observed colligative property}}{\text{calculated}} = \dfrac{\text{normal molar mass}}{\text{abnormal molar mass}}$. Dissociation: $\alpha = \dfrac{i-1}{n-1}$; association: $\alpha = \dfrac{1-i}{1-\frac{1}{n}}$.

1B Diagrams and graphs to practise

Vapour pressure vs mole fraction (ideal, positive and negative deviation)



ideal positive deviation negative deviation

Also practise: vapour pressure vs temperature curves showing ΔT_b and ΔT_f; osmosis/reverse osmosis set-up.

1C Exercise-wise numerical patterns

Exercises 1.1–1.39 (39 questions, 6 patterns)

Pattern A: Concentration terms (%, x, m, M, ppm)

Solved (Q1.4): take 1 L = 1504 g solution; $HNO_3 = 0.68 \times 1504 = 1022.7$ g $= \frac{1022.7}{63} = 16.2$ mol. Molarity = 16.2 M.

Pattern B: Henry's law

$p = K_H x$ (solubility $\propto$ pressure). **Solved (Q1.13):** mass dissolved $\propto p$: $p = 1 \times \dfrac{5.00 \times 10^{-2}}{6.56 \times 10^{-3}} = 7.6$ bar.

Pattern C: Raoult's law (volatile components, RLVP)

Solved (Q1.16): heptane $\frac{26}{100} = 0.260$ mol, octane $\frac{35}{114} = 0.307$ mol; $x_{hep} = 0.459$.
$p = 0.459(105.2) + 0.541(46.8) = 73.6$ kPa.

Pattern D: ΔT_b, ΔT_f (find K, M or new freezing point)

Solved (Q1.20): sucrose: $m = \frac{5/342}{0.095} = 0.154$; $\Delta T_f = 2.15$ K $\Rightarrow K_f = 13.97$.
Glucose: $m = \frac{5/180}{0.095} = 0.292$; $\Delta T_f = 13.97 \times 0.292 = 4.08$ K; freezing point = 269.07 K.

Pattern E: Osmotic pressure

Solved (Q1.22): $\pi \propto C$ at fixed T: $C_1 = \frac{36/180}{1} = 0.2$ M, so $C_2 = 0.2 \times \frac{1.52}{4.98} = 0.061$ M.

Pattern F: van't Hoff factor (dissociation / association)

Solved (Q1.33): $m = \frac{19.5/78}{0.5} = 0.5$; expected $\Delta T_f = 0.93$ K, observed 1.00 K: $i = 1.075$, $\alpha = 0.075$.
$K_a = \dfrac{c\alpha^2}{1-\alpha} = \dfrac{0.5(0.075)^2}{0.925} = 3.07 \times 10^{-3}$.

Pattern A: Q1.1–1.9, 1.26, 1.28, 1.29, 1.30. Pattern B: Q1.11–1.13, 1.35, 1.39. Pattern C: Q1.14–1.19, 1.34, 1.36–1.38. Pattern D: Q1.20, 1.21. Pattern E: Q1.22. Pattern F: Q1.27, 1.31–1.33. Theory (interactions, solubility): Q1.10, 1.23–1.25.

1D Conceptual / reasoning questions

- **Gases less soluble at higher temperature** → Dissolution is exothermic; Le Chatelier.
- **Divers get bends** → High-pressure N_2 dissolves in blood (Henry); bubbles on surfacing.
- **Ethanol + water: which deviation?** → Positive: weaker H-bonds in the mixture; $\Delta_{mix}H > 0$, volume increases.
- **Chloroform + acetone?** → Negative: new H-bonds; $\Delta_{mix}H < 0$; maximum-boiling azeotrope.
- **Salt spread on icy roads** → Lowers the freezing point of water.
- **Why osmotic pressure for proteins?** → Large value at room temperature; no heating that could denature.
- **Order of ΔT_f: CH_3COOH < CCl_3COOH < CF_3COOH** → Stronger acid dissociates more: larger i.
- **Molality preferred to molarity** → Independent of temperature (mass does not change).

1E Exam tips

Common mistakes and unit traps

- Molality uses kg of *solvent*; molarity uses L of *solution* (use density).
- $R = 0.083$ L bar K^{-1} mol^{-1} for π in bar; 8.314 J for SI. Volume in L with 0.083.
- Forgetting i for electrolytes (NaCl $i = 2$, $CaCl_2$ $i = 3$, $K_4[Fe(CN)_6]$ $i = 5$).
- Temperature in K; ΔT same in K and °C.

Question types: 1-mark: definitions, deviation type, ideal solution conditions; 2–3 marks: concentration, Henry, Raoult numericals; 5 marks: colligative property with molar mass or i.

1F JEE extras

Useful results

- Vapour composition: $y_1 = \dfrac{x_1 p_1^\circ}{p_{total}}$; vapour is richer in the more volatile component.
- $K_b = \dfrac{RT_b^2 M_1}{1000\,\Delta_{vap}H}$. Isotonic solutions: $C_1 i_1 = C_2 i_2$.
- Weak electrolyte $AB \rightleftharpoons A^+ + B^-$: $i = 1 + \alpha$; dimerisation (e.g. acetic acid in benzene): $i = 1 - \frac{\alpha}{2}$.

Example (Q1.38): $x_B = \frac{80/78}{80/78+100/92} = 0.486$, $p_B = 24.6$, $p_T = 16.5$ mm Hg; $y_B = \frac{24.6}{41.1} = 0.60$.

1G Pattern spotter

If you see…	Do this	See
Concentration		
% and density given, molarity asked	Take 1 L of solution, mass = $1000 \times d$	Ex A
Molality / mole fraction	Moles of each; kg of solvent	Ex A
Laws		
Gas mass or pressure in liquid	Henry: $p = K_H x$	Ex B
Two volatile liquids	$p = x_1 p_1^\circ + x_2 p_2^\circ$	Ex C
Non-volatile solute, vapour pressure	RLVP = x_2	Ex C
Colligative properties		
Change of boiling/freezing point	$\Delta T = iKm$	Ex D
Osmotic pressure	$\pi = iCRT$	Ex E
Observed $\neq$ calculated	i, then α (and K_a)	Ex F

Chapter 2 Electrochemistry

2A Derivations (board favourites)

1. Nernst equation for a cell

For $aA + bB \rightarrow cC + dD$ (n electrons): $E_{cell} = E^\circ_{cell} - \dfrac{2.303RT}{nF}\log\dfrac{[C]^c[D]^d}{[A]^a[B]^b}$.

At 298 K: $\boxed{E_{cell} = E^\circ_{cell} - \dfrac{0.059}{n}\log Q}$. Daniell cell: $E = 1.1 - \dfrac{0.059}{2}\log\dfrac{[Zn^{2+}]}{[Cu^{2+}]}$.

2. E°, ΔG° and K

At equilibrium $E_{cell} = 0$: $E^\circ = \dfrac{2.303RT}{nF}\log K_c$; $\boxed{\Delta_r G^\circ = -nFE^\circ_{cell} = -2.303RT\log K_c}$.

3. Kohlrausch's law and Λ_m

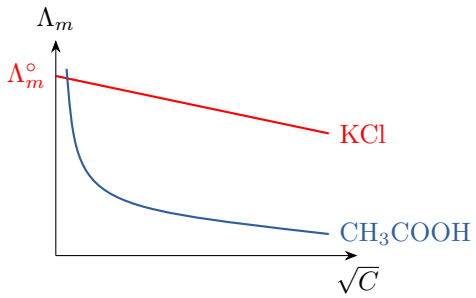
$\Lambda_m = \dfrac{\kappa \times 1000}{C}$ (S cm^2 mol^{-1}). Strong electrolyte: $\Lambda_m = \Lambda^\circ_m - A\sqrt{C}$.

$\Lambda^\circ_m = \nu_+\lambda^\circ_+ + \nu_-\lambda^\circ_-$. Weak electrolyte: $\alpha = \dfrac{\Lambda_m}{\Lambda^\circ_m}$, $K_a = \dfrac{C\alpha^2}{1-\alpha}$.

$\Lambda^\circ_m(CH_3COOH) = \Lambda^\circ(CH_3COONa) + \Lambda^\circ(HCl) - \Lambda^\circ(NaCl)$.

2B Diagrams and graphs to practise

Λ_m vs $\sqrt{C}$: strong (KCl) and weak (CH_3COOH) electrolytes



Also practise: Daniell cell (salt bridge, anode − / cathode +, electron flow), standard hydrogen electrode, lead storage battery, H_2–O_2 fuel cell, rusting.

2C Exercise-wise numerical patterns

Exercises 2.1–2.18 (18 questions, 6 patterns)

Pattern A: Electrochemical series (reducing power, displacement, feasibility)

Lower E° = stronger reducing agent. **Solved (Q2.2):** E°: Ag 0.80, Hg 0.79, Cr -0.74, Mg -2.37, K -2.93.
Increasing reducing power: Ag < Hg < Cr < Mg < K.

Pattern B: E°_{cell}, ΔG°, K

Solved (Q2.4(i)): $E^\circ = E^\circ_{cathode} - E^\circ_{anode} = -0.40 - (-0.74) = 0.34$ V; $n = 6$.
$\Delta_r G^\circ = -6 \times 96487 \times 0.34 = -196.8$ kJ mol^{-1}; $\log K = \frac{6\times 0.34}{0.059} = 34.6$, $K = 3.8 \times 10^{34}$.

Pattern C: Nernst equation

Solved (Q2.5(i)): Mg|Mg^{2+}(0.001)‖Cu^{2+}(0.0001)|Cu: $E^\circ = 0.34 + 2.37 = 2.71$ V.
$E = 2.71 - \frac{0.059}{2} \log \frac{0.001}{0.0001} = 2.71 - 0.03 = 2.68$ V.

Pattern D: Conductivity, molar conductivity, α, K_a

Solved (Q2.11): $\Lambda_m = \frac{7.896\times 10^{-5}\times 1000}{0.00241} = 32.8$ S cm^2 mol^{-1}; $\alpha = \frac{32.8}{390.5} = 0.084$.
$K_a = \frac{C\alpha^2}{1-\alpha} = 1.85 \times 10^{-5}$. **(Q2.9):** cell constant $G^* = \kappa R = 0.146 \times 10^{-3} \times 1500 = 0.219$ cm^{-1}.

Pattern E: Faraday's laws

Solved (Q2.15): $Q = 5 \times 20 \times 60 = 6000$ C; Ni^{2+} needs 2 F per mol: $m = \frac{58.7\times 6000}{2\times 96487} = 1.82$ g.
Series cells (Q2.16): same charge, so masses are in the ratio of equivalent weights (M/n).

Pattern F: Products of electrolysis

Solved (Q2.18): (i) $AgNO_3$, Ag electrodes: Ag deposits at cathode, Ag anode dissolves. (ii) Pt electrodes: Ag at cathode, O_2 at anode. (iii) dil. H_2SO_4: H_2 at cathode, O_2 at anode. (iv) $CuCl_2$: Cu at cathode, Cl_2 at anode.

Pattern A: Q2.1, 2.2, 2.17. Pattern B: Q2.3, 2.4, 2.6. Pattern C: Q2.5. Pattern D: Q2.7–2.11. Pattern E: Q2.12–2.16. Pattern F: Q2.18.

2D Conceptual / reasoning questions

- **Can $CuSO_4$ be stored in a zinc pot?** → No; Zn (-0.76 V) displaces Cu.
- **Conductivity decreases on dilution but Λ_m increases** → Fewer ions per cm^3; but Λ_m counts all ions of 1 mol (more dissociation, less attraction).
- **Λ_m° of a weak electrolyte** → Cannot be found by extrapolation; use Kohlrausch's law.
- **Function of salt bridge** → Completes the circuit and keeps solutions neutral.
- **Rusting** → Electrochemical: Fe is the anode, O_2 reduced at cathode; prevented by galvanising/sacrificial zinc.
- **Advantage of fuel cells** → High efficiency ($\sim 70\%$), no pollution.

2E Exam tips

Common mistakes and unit traps

- $E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$ (both as reduction potentials); never multiply E° by coefficients.
- In Nernst, Q = products over reactants; solids and pure liquids are 1; n = electrons transferred.
- Λ_m with κ in S cm^{-1} and C in mol L^{-1}: multiply by 1000.
- ΔG in J; convert to kJ. Time in seconds for $Q = It$.

Question types: 1-mark: cell notation, units, which is anode; 3 marks: Nernst, ΔG, K, Faraday numericals; 5 marks: Kohlrausch + α, batteries and corrosion.

2F JEE extras

Useful results

- Concentration cell: $E = \frac{0.059}{n} \log \frac{C_2}{C_1}$ ($E^\circ = 0$).
- Hydrogen electrode: $E = -0.059\,\text{pH}$.
- Combining half-cells: add ΔG (nE°), not E°: $E^\circ_3 = \frac{n_1 E^\circ_1 + n_2 E^\circ_2}{n_3}$.

Example: H_2 electrode in pH 10: $E = -0.59$ V.

2G Pattern spotter

If you see...	Do this	See
Cells		
Which metal displaces / reduces more	Lower E°	Ex A
E°_{cell}, ΔG°, K	Cathode − anode; $-nFE$; $\log K = \frac{nE}{0.059}$	Ex B
Non-standard concentrations	Nernst equation	Ex C
Is a reaction feasible?	$E^\circ_{cell} > 0$	Ex A
Conductance and electrolysis		
κ given, Λ_m asked	$\frac{1000\kappa}{C}$	Ex D
Weak acid α, K_a	$\Lambda_m/\Lambda_m^\circ$	Ex D

If you see…	Do this	See
Mass deposited / charge needed	$m = \frac{MIt}{nF}$	Ex E
Products of electrolysis	Compare E° with water; electrode material	Ex F

Chapter 3 Chemical Kinetics

3A Derivations (board favourites)

1. Integrated rate law: zero order

$$-\frac{d[\mathrm{R}]}{dt} = k \Rightarrow [\mathrm{R}] = [\mathrm{R}]_0 - kt, \quad \boxed{k = \frac{[\mathrm{R}]_0 - [\mathrm{R}]}{t}}, \quad t_{1/2} = \frac{[\mathrm{R}]_0}{2k}.$$

2. Integrated rate law: first order

$$-\frac{d[\mathrm{R}]}{dt} = k[\mathrm{R}] \Rightarrow \ln\frac{[\mathrm{R}]_0}{[\mathrm{R}]} = kt, \quad \boxed{k = \frac{2.303}{t}\log\frac{[\mathrm{R}]_0}{[\mathrm{R}]}}; \quad t_{1/2} = \frac{0.693}{k} \text{ (independent of } [\mathrm{R}]_0).$$

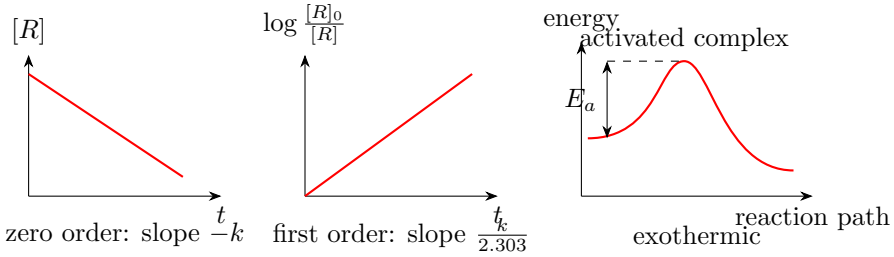
Gas phase $A(g) \rightarrow B(g) + C(g)$: $k = \dfrac{2.303}{t}\log\dfrac{p_i}{2p_i - p_t}$.

3. Arrhenius equation

$k = Ae^{-E_a/RT}$; $\log\dfrac{k_2}{k_1} = \dfrac{E_a}{2.303R}\left(\dfrac{1}{T_1} - \dfrac{1}{T_2}\right)$. Plot of $\ln k$ vs $\frac{1}{T}$: slope $-\frac{E_a}{R}$.

3B Diagrams and graphs to practise

Graphs (zero order, first order, energy profile)



3C Exercise-wise numerical patterns

Exercises 3.1–3.30 (30 questions, 6 patterns)

Pattern A: Order, units of k, rate from the rate law

Units of k: $(\text{mol L}^{-1})^{1-n}\ \text{s}^{-1}$. **Solved (Q3.1(iii)):** rate $= k[\text{CH}_3\text{CHO}]^{3/2}$: order $\frac{3}{2}$, units $\text{mol}^{-1/2}\ \text{L}^{1/2}\ \text{s}^{-1}$.
(Q3.6): second order: doubling gives 4× rate; halving gives $\frac{1}{4}$.

Pattern B: Initial-rate method

Solved (Q3.11): I → IV: [A] ×4, rate ×4: first order in A. II → III: [B] ×2, rate ×4: second order in B.
Rate $= k[\text{A}][\text{B}]^2$; $k = \frac{6.0\times10^{-3}}{0.1\times0.01} = 6.0\ \text{mol}^{-2}\ \text{L}^2\ \text{min}^{-1}$.

Pattern C: First-order calculations (t, k, $t_{1/2}$, amount left)

Solved (Q3.14): $k = \frac{0.693}{5730} = 1.21\times10^{-4}\ \text{y}^{-1}$; $t = \frac{2.303}{k}\log\frac{100}{80} = 1845$ years.
(Q3.18): $t_{99\%} = \frac{2.303}{k}\log 100 = \frac{4.606}{k}$, $t_{90\%} = \frac{2.303}{k}\log 10 = \frac{2.303}{k}$: ratio 2.

Pattern D: Pressure data (gas-phase first order)

Solved (Q3.21): $k = \frac{2.303}{100}\log\frac{0.5}{2(0.5)-0.6} = 2.23\times10^{-3}\ \text{s}^{-1}$.
At $p_t = 0.65$: $p_{\text{SO}_2\text{Cl}_2} = 1.0 - 0.65 = 0.35$ atm; rate $= kp = 7.8\times10^{-4}$ atm s^{-1}.

Pattern E: Arrhenius (E_a, A, k at another T)

Solved (Q3.30): $\log 4 = \dfrac{E_a}{2.303\times8.314}\left(\dfrac{1}{293}-\dfrac{1}{313}\right) \Rightarrow E_a = 52.9\ \text{kJ mol}^{-1}$.

Pattern F: Average rate from data

Solved (Q3.8): $r_{av} = -\frac{\Delta[A]}{\Delta t} = -\frac{0.17-0.31}{60-30} = 4.67\times10^{-3}\ \text{mol L}^{-1}\ \text{s}^{-1}$.

Pattern A: Q3.1–3.4, 3.6, 3.9. Pattern B: Q3.10, 3.11, 3.12. Pattern C: Q3.13–3.19, 3.24, 3.25. Pattern D: Q3.20, 3.21. Pattern E: Q3.7, 3.22, 3.23, 3.26–3.30. Pattern F: Q3.5, 3.8.

3D Conceptual / reasoning questions

- **Order vs molecularity** → Order is experimental, can be 0 or fractional; molecularity is theoretical, a whole number for an elementary step.
- **Pseudo first order** → One reactant in large excess (e.g. hydrolysis of ester in water).
- **Why does rate increase with temperature?** → More molecules exceed E_a (Maxwell distribution); a 10 K rise often doubles the rate.
- **Effect of catalyst** → Lowers E_a by a new pathway; does not change ΔG or equilibrium.
- **Why not all collisions are effective?** → Need energy $\geq E_a$ and proper orientation.

3E Exam tips

Common mistakes and unit traps

- Rate of reaction $= -\frac{1}{a}\frac{d[\mathrm{A}]}{dt}$ (divide by the coefficient).
- Use log (base 10) with 2.303. Keep time units consistent with k.
- In Arrhenius, $R = 8.314$ J, so E_a comes in J; T in kelvin.
- Order comes from experiment, not from the balanced equation.

Question types: 1-mark: units, order, $t_{1/2}$; 3 marks: initial-rate tables, first-order numericals; 5 marks: derivation of integrated rate law with a numerical, Arrhenius.

3F JEE extras

Useful results

- nth order: $t_{1/2} \propto [\mathrm{R}]_0^{1-n}$; second order: $\frac{1}{[\mathrm{R}]} - \frac{1}{[\mathrm{R}]_0} = kt$.
- After n half-lives: $[\mathrm{R}] = \frac{[\mathrm{R}]_0}{2^n}$ (Q3.16: $\frac{1}{16}$ needs $4t_{1/2} = 4 \times \frac{0.693}{60} = 0.046$ s).
- Temperature coefficient $\frac{k_{T+10}}{k_T} \approx 2$–3.

3G Pattern spotter

If you see...	Do this	See
Rate laws		
Rate law given, order or units asked	Sum of powers; $(\mathrm{mol\ L^{-1}})^{1-n}\mathrm{s}^{-1}$	Ex A
Table of initial rates	Compare experiments with one change	Ex B
Concentration–time table	Average rate $= -\Delta[A]/\Delta t$	Ex F
Integrated laws		
First order, find t or amount left	$k = \frac{2.303}{t}\log\frac{[R]_0}{[R]}$	Ex C
Half-life	$0.693/k$	Ex C
Total pressure data	$\log\frac{p_i}{2p_i - p_t}$	Ex D
Temperature		
Two temperatures, two k	Arrhenius two-point form	Ex E

If you see...	Do this	See
$k = Ae^{-E_a/RT}$ given	Compare exponents	Ex E

Chapter 4 The d- and f-Block Elements

4A Give-reasons bank

Each reason is the key phrase examiners look for. Learn the phrase, then the example.

d-block: general properties

Statement	Reason
Zn, Cd, Hg are not regarded as transition elements	Full d^{10} in the atom and in their common ions
Cu and Ag are transition elements despite d^{10} atoms	Cu^{2+} ($3d^9$) and Ag^{2+} ($4d^9$) have incomplete d subshells
High enthalpies of atomisation	Many unpaired d electrons $\Rightarrow$ strong metallic bonding
Zn has the lowest enthalpy of atomisation	No unpaired electrons ($3d^{10}4s^2$) $\Rightarrow$ weak metallic bonding
Transition metals are paramagnetic	Unpaired d electrons
Coloured compounds	d–d transitions absorb visible light; we see the complementary colour
Sc^{3+}, Ti^{4+}, Zn^{2+}, Cu^{+} are colourless	d^0 or d^{10}: no d–d transition possible
Good catalysts	Variable oxidation states and ability to form complexes / surface intermediates
Form many complexes	Small ions, high charge, vacant d orbitals
Form interstitial compounds	Small atoms (H, C, N) fit into holes of the metal lattice
Form alloys readily	Similar atomic radii (within about 15%)
Variable oxidation states	$(n-1)d$ and ns electrons are close in energy, so both take part
Zr and Hf have almost the same radius	Lanthanoid contraction cancels the expected size increase

Oxidation states and stability

Statement	Reason
Mn shows the most oxidation states (+2 to +7)	It has the most unpaired electrons to use ($3d^5 4s^2$)

Sc shows only +3	Sc^{3+} has the stable noble gas configuration
Mn^{2+} more stable than Fe^{2+} to oxidation	Mn^{2+} is d^5 (half-filled); Fe^{2+} (d^6) $\rightarrow$ Fe^{3+} (d^5) easily
Highest oxidation state only in oxides and fluorides	O and F are small and highly electronegative; O forms multiple bonds
Highest oxidation state in oxoanions (MnO_4^-, CrO_4^{2-})	Multiple bonding to O stabilises the high state
Lower oxide basic, higher oxide acidic	Higher oxidation state $\Rightarrow$ more covalent, more acidic
Cr^{2+} reducing, Mn^{3+} oxidising (both d^4)	$Cr^{2+} \rightarrow Cr^{3+}$ gives d^3 (half-filled t_{2g}); $Mn^{3+} \rightarrow Mn^{2+}$ gives d^5
$E^\circ(Mn^{3+}/Mn^{2+})$ very positive	Very high IE_3 of Mn (breaking d^5)
$E^\circ(Cu^{2+}/Cu)$ positive (+0.34 V)	High $\Delta_a H$ and IE_2 are not balanced by $\Delta_{hyd} H$
Cu^+ unstable in water	Disproportionates; high $\Delta_{hyd} H$ of Cu^{2+} more than balances IE_2
Co(II) stable in water but oxidised with complexing reagents	Strong-field ligands stabilise Co(III) (t_{2g}^6)
d^1 ions are unstable	Losing the single electron gives a stable d^0 configuration
Cu shows +1 most often in the 3d series	Cu^+ is $3d^{10}$ (stable full d)

f-block

Statement	Reason
Lanthanoid contraction	Poor shielding by $4f$ electrons; effective nuclear charge rises
Actinoid contraction greater than lanthanoid	$5f$ electrons shield even more poorly than $4f$
Ce^{4+} is a good oxidant	Prefers +3; but reaction with water is slow, so useful in analysis
Eu^{2+}, Yb^{2+} are reducing	They go to the common +3 state
Actinoids show more oxidation states	$5f$, $6d$, $7s$ are close in energy
Actinoid chemistry is harder to study	Radioactive, short half-lives, many oxidation states

4B Exercise-wise solution patterns

Exercises 4.1–4.30 (30 questions, 5 patterns)

Pattern A: Electronic configuration, unpaired electrons, magnetic moment

Spot: "write the configuration", "number of unpaired electrons", "spin-only moment". **Method:** write the atom, remove 4*s* first, count unpaired *d* electrons, use $\mu = \sqrt{n(n+2)}$.

Solved (Q4.24): Mn^{3+} $3d^4$: 4 unpaired; Cr^{3+} $3d^3$: 3; V^{3+} $3d^2$: 2; Ti^{3+} $3d^1$: 1.
Most stable in water: **Cr^{3+}**, because of its half-filled t_{2g}^3 level.

Pattern B: Stability of an oxidation state (configuration or E° data)

Spot: "more stable", "oxidising / reducing", a table of E° values. **Method:** compare the *d* configuration before and after; for E°, more positive M^{3+}/M^{2+} means M^{3+} is easily reduced (unstable); more negative M^{2+}/M means the metal is easily oxidised.

Solved (Q4.17): (i) $E^\circ(Fe^{3+}/Fe^{2+}) = +0.8$ V is much lower than Mn^{3+}/Mn^{2+} (+1.5 V), so Fe^{3+} is more stable than Mn^{3+}; Cr^{3+}/Cr^{2+} is −0.4 V, so Cr^{3+} is the most stable.
(ii) $E^\circ(M^{2+}/M)$: Mn −1.2, Cr −0.9, Fe −0.4. Ease of oxidation: **Mn** > **Cr** > **Fe**, so iron is the hardest to oxidise.

Pattern C: Give reasons for general properties

Spot: "explain", "account for", "predict which are coloured". **Method:** name the cause in one phrase (Part A), then give an example.

Solved (Q4.18): coloured if the ion has a partly filled *d* subshell.
Ti^{3+} (d^1), V^{3+} (d^2), Mn^{2+} (d^5), Fe^{3+} (d^5), Co^{2+} (d^7): **coloured**. Cu^+ (d^{10}), Sc^{3+} (d^0): **colourless**.

Pattern D: $K_2Cr_2O_7$ and $KMnO_4$: preparation and ionic equations

Spot: "describe the preparation", "write the ionic equation", "disproportionation". **Method:** write the oxidant half-reaction, write the reductant half-reaction, multiply so the electrons match, add.

Solved (Q4.15): $Cr_2O_7^{2-} + 14\,H^+ + 6\,e^- \longrightarrow 2\,Cr^{3+} + 7\,H_2O$
(i) $6\,I^- \longrightarrow 3\,I_2 + 6\,e^-$: $Cr_2O_7^{2-} + 14\,H^+ + 6\,I^- \longrightarrow 2\,Cr^{3+} + 3\,I_2 + 7\,H_2O$
(ii) $6\,Fe^{2+} \longrightarrow 6\,Fe^{3+} + 6\,e^-$: $Cr_2O_7^{2-} + 14\,H^+ + 6\,Fe^{2+} \longrightarrow 2\,Cr^{3+} + 6\,Fe^{3+} + 7\,H_2O$
(iii) $3\,H_2S \longrightarrow 6\,H^+ + 3\,S + 6\,e^-$: $Cr_2O_7^{2-} + 8\,H^+ + 3\,H_2S \longrightarrow 2\,Cr^{3+} + 3\,S + 7\,H_2O$

Pattern E: Lanthanoids and actinoids

Spot: "lanthanoid contraction", "compare actinoids with lanthanoids", "which atomic numbers are inner transition".

Solved (Q4.28): inner transition elements have $Z = 58$–71 (lanthanoids) or 90–103 (actinoids).
From $29, 59, 74, 95, 102, 104$: **59** (Pr), **95** (Am), **102** (No). (29 Cu, 74 W, 104 Rf are d-block.)

Pattern A: Q4.1, 4.5, 4.9, 4.24, 4.30. Pattern B: Q4.2, 4.3, 4.4, 4.13, 4.17, 4.19, 4.21, 4.23. Pattern C: Q4.8, 4.11, 4.12, 4.18, 4.25. Pattern D: Q4.6, 4.14, 4.15, 4.16, 4.22, 4.26. Pattern E: Q4.7, 4.10, 4.20, 4.27, 4.28, 4.29.

Intext Questions 4.1–4.10 (10 questions, 3 patterns)

Pattern A: Configuration and magnetic moment

Solved (Q4.8): $Z = 27$ is Co: $3d^7 4s^2$, so M^{2+} is $3d^7$ with 3 unpaired electrons. $\mu = \sqrt{3(3+2)} = \sqrt{15} = 3.87$ BM.

Pattern B: Trends and their exceptions

Solved (Q4.5): IE does not rise smoothly because of the extra stability of d^5 and d^{10}. IE_1 of Cr is low (losing $4s^1$ keeps d^5); IE_2 of Cr and Cu is very high (breaking d^5 / d^{10}); IE_2 of Zn is low (gives d^{10}).

Pattern C: Stability and redox

Solved (Q4.7): Cr^{2+} is the stronger reducing agent: $Cr^{2+} \rightarrow Cr^{3+}$ ($d^4 \rightarrow d^3$, half-filled t_{2g}) is favoured, while $Fe^{2+} \rightarrow Fe^{3+}$ ($d^6 \rightarrow d^5$) is less favourable in comparison. (E° $Cr^{3+}/Cr^{2+} = -0.41$ V vs $Fe^{3+}/Fe^{2+} = +0.77$ V.)

Pattern A: Q4.1, 4.8. Pattern B: Q4.2, 4.3, 4.4, 4.5, 4.10. Pattern C: Q4.6, 4.7, 4.9.

4C Exam tips

Which approach? (read the question's key words)

1. **"Configuration", "unpaired", "magnetic", "coloured"** → write the ion's d^n first; everything follows from n.
2. **"More stable", "oxidising", "reducing"** → what configuration does it go to? Half-filled / full / d^0 / t_{2g}^3 is the answer most of the time.
3. **E° data given** → more positive M^{3+}/M^{2+}: M^{3+} is a strong oxidant; more negative M^{2+}/M: metal easily oxidised.
4. **"Explain / give reasons"** → key phrase from Part A plus one example.
5. **"Describe preparation" / "ionic equation"** → balanced equations with conditions; balance by electrons.
6. **Lanthanoid / actinoid comparison** → use the four headings: configuration, size, oxidation states, reactivity.

Common mistakes that lose marks

- Removing $3d$ electrons before $4s$ when forming ions.
- Writing Cr as $3d^4 4s^2$ or Cu as $3d^9 4s^2$.
- Calling Zn a transition element, or saying a d^{10} ion is coloured.
- Unbalanced ionic equations: check atoms *and* charge. Dichromate always needs $14H^+$ per $Cr_2O_7^{2-}$ with $6e^-$.
- Mixing up the three $KMnO_4$ products: Mn^{2+} (acid), MnO_2 (neutral/weakly alkaline), MnO_4^{2-} (strongly alkaline).
- Saying the oxidation state of Cr changes between chromate and dichromate (it is +6 in both).
- Forgetting the *consequences* when asked about lanthanoid contraction.

Smart exam habits and question types

- Answer "give reasons" in one or two lines: *cause* + *example*. Long paragraphs earn no extra marks.
- Write the d^n configuration next to each ion; it shows your reasoning and earns step marks.

- **1-mark / MCQ:** configurations, colour or magnetic behaviour of an ion, oxidation state of Cr/Mn in a compound, assertion–reason on lanthanoid contraction.
- **2–3 marks:** give-reasons sets, ionic equations of $KMnO_4$/$K_2Cr_2O_7$, preparation steps.
- **5 marks / case-based:** comparison of lanthanoids and actinoids, E° data interpretation, mixed give-reasons.

4D JEE extras

J1. Redox titration stoichiometry

Electrons accepted: MnO_4^- 5 (acid), $Cr_2O_7^{2-}$ 6. Electrons given: Fe^{2+} 1, $C_2O_4^{2-}$ 2, I^- 1.

Example: volume of 0.02 M $KMnO_4$ needed for 20 mL of 0.1 M $FeSO_4$ in acid.
Moles $Fe^{2+} = 0.002$; MnO_4^- needed $= \frac{0.002}{5} = 0.0004$ mol; volume $= \frac{0.0004}{0.02} = 0.02$ L $=$ **20 mL**.

J2. Magnetic moment backwards

Example: an ion of a 3d metal shows $\mu = 4.90$ BM. Then $n(n+2) = 24 \Rightarrow n = 4$: the ion is d^4 or d^6 (high spin), e.g. Cr^{2+}, Mn^{3+} or Fe^{2+}.

J3. Using E° to predict whether a reaction happens

$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}} > 0$ means the reaction is feasible.
Example: can Fe^{3+} oxidise I^-? $E^\circ(Fe^{3+}/Fe^{2+}) = 0.77$ V, $E^\circ(I_2/I^-) = 0.54$ V: $0.77 - 0.54 = 0.23$ V > 0, so yes: $2\,Fe^{3+} + 2\,I^- \longrightarrow 2\,Fe^{2+} + I_2$.

J4. Tests worth knowing

- **Chromyl chloride test** for Cl^-: heat with $K_2Cr_2O_7$ and conc. H_2SO_4; red-brown CrO_2Cl_2 vapour; passed into NaOH gives yellow Na_2CrO_4.
- **CrO_5**: acidified dichromate + H_2O_2 gives a blue colour in ether layer; Cr is +6 (two peroxo groups).
- $KMnO_4$ is **self-indicator**: the end point is the first permanent pink.

4E Pattern spotter

If you see...	Do this	See
Configuration and magnetism		
Configuration of an ion	Atom first; remove $4s$ before $3d$	Ex A
Number of unpaired electrons	Fill d^n by Hund's rule	Ex A
Spin-only moment	$\mu = \sqrt{n(n+2)}$	Intext A
Coloured or not?	Partly filled d: coloured; d^0, d^{10}: colourless	Ex C
Stability and redox		
"Which is more stable?"	Compare configurations reached (half-filled, full, d^0, t_{2g}^3)	Ex B
$E^\circ(M^{3+}/M^{2+})$ data	More positive: M^{3+} strong oxidant, M^{2+} stable	Ex B
$E^\circ(M^{2+}/M)$ data	More negative: metal more easily oxidised	Ex B
d^4 ions Cr^{2+}, Mn^{3+}	Cr^{2+} reducing ($\rightarrow d^3$), Mn^{3+} oxidising ($\rightarrow d^5$)	4A
Irregular trend	Point to the d^5 / d^{10} exception	Intext B
Highest oxidation state	Oxides, fluorides, oxoanions	4A
Compounds		
Preparation of $K_2Cr_2O_7$ / $KMnO_4$	Three steps from ore, with equations	Ex D
Ionic equation with a reductant	Balance half-reactions by electrons	Ex D
Effect of pH on dichromate	CrO_4^{2-} (yellow) $\rightleftharpoons$ $Cr_2O_7^{2-}$ (orange)	Ex D
Disproportionation	One state $\rightarrow$ higher + lower, e.g. MnO_4^{2-}	Ex D
f-block		
Lanthanoid contraction	Poor $4f$ shielding; Zr $\approx$ Hf, basicity decreases	Ex E
Is this Z inner transition?	58–71 or 90–103	Ex E
Compare Ln and An	Configuration, size, oxidation states, reactivity	Ex E
JEE extras		
Titration volume	Match electrons: MnO_4^- 5, $Cr_2O_7^{2-}$ 6	J1
Given μ, find ion	$n(n+2) = \mu^2$	J2
Will the reaction happen?	$E^\circ_{\text{cell}} > 0$	J3

Chapter 5 Coordination Compounds

5A Give-reasons bank

Key reasons

Statement	Reason
Double salt (Mohr's salt) gives Fe^{2+} test; complex $[Cu(NH_3)_4]^{2+}$ does not give Cu^{2+} test	Double salts dissociate fully; complexes keep their coordination entity intact
CN^- is a strong-field ligand	Large Δ_o; forces pairing (low spin)
$[Ni(CN)_4]^{2-}$ diamagnetic, square planar	CN^- pairs $3d$ electrons; dsp^2, no unpaired electron
$[NiCl_4]^{2-}$ paramagnetic, tetrahedral	Cl^- is weak field; sp^3, 2 unpaired electrons
$[Ti(H_2O)_6]^{3+}$ is violet	d–d transition $t_{2g}^1 \rightarrow e_g^1$ absorbs green-yellow light
Anhydrous $CuSO_4$ white, hydrated blue	No ligand field without water, so no d–d transition
$[Ni(CN)_4]^{2-}$ colourless, $[Ni(H_2O)_6]^{2+}$ green	No d–d absorption in the visible for the strong-field square planar complex
Chelates are more stable	Chelate effect: ring formation (entropy increase)
No CuS precipitate from Cu^{2+} + excess KCN	Stable $[Cu(CN)_4]^{3-}$ forms; free Cu^{2+} is negligible
Metal carbonyls are stable	Synergic bonding: σ donation by CO and π back bonding from metal
$[Fe(CN)_6]^{4-}$ and $[Fe(H_2O)_6]^{2+}$ differ in colour	Different Δ_o (strong vs weak field) absorb different wavelengths

5B Exercise-wise solution patterns

Exercises 5.1–5.31 (31 questions, 6 patterns)

Pattern A: IUPAC names and formulas

Rules: cation first; ligands alphabetically (prefixes di-, tri- ignored) before the metal; anionic ligands end in -o (chlorido, cyanido, oxalato); anionic complex: metal ends in -ate (ferrate, cuprate, argentate, aurate); oxidation state in Roman numerals.

Solved (Q5.7(iv)): $[Co(NH_3)_4Cl(NO_2)]Cl$: tetraamminechloridonitrito-N-cobalt(III) chloride.
(Q5.6(vii)): Potassium tri(oxalato)chromate(III): $K_3[Cr(C_2O_4)_3]$.

Pattern B: Oxidation state, coordination number, d-configuration

Solved (Q5.5(i)): $[Co(H_2O)(CN)(en)_2]^{2+}$: $x + 0 - 1 + 0 = +2 \Rightarrow x = +3$; CN $= 1 + 1 + 2 \times 2 = 6$.
(Q5.23(i)): $K_3[Co(C_2O_4)_3]$: Co^{3+}, d^6 (t_{2g}^6, low spin with oxalate in NCERT), CN 6.

Pattern C: Isomerism

Solved (Q5.9): $[Cr(C_2O_4)_3]^{3-}$: no geometrical isomers (but optical). $[Co(NH_3)_3Cl_3]$: 2 (fac and mer).
(Q5.12): $[Pt(NH_3)(Br)(Cl)(py)]$ square planar: 3 geometrical isomers; none optically active (planar).

Pattern D: Valence bond theory

Solved (Q5.15(i),(ii)): $[Fe(CN)_6]^{4-}$: Fe^{2+} $3d^6$, CN^- strong: pairs electrons, d^2sp^3, inner orbital, octahedral, diamagnetic.
$[FeF_6]^{3-}$: Fe^{3+} $3d^5$, F^- weak: sp^3d^2, outer orbital, octahedral, 5 unpaired ($\mu = 5.92$ BM).

Pattern E: Crystal field theory (splitting, colour, magnetism)

Spectrochemical series: $I^- < Br^- < SCN^- < Cl^- < S^{2-} < F^- < OH^- < C_2O_4^{2-} < H_2O < NCS^- < edta^{4-} < NH_3 < en < CN^- < CO$.

Solved (Q5.31): larger $\Delta_o \Rightarrow$ shorter wavelength absorbed. NO_2^- > NH_3 > H_2O, so λ: $[Ni(NO_2)_6]^{4-} < [Ni(NH_3)_6]^{2+} < [Ni(H_2O)_6]^{2+}$.

Pattern F: MCQs (ions in solution, magnetic moment, stability)

Solved (Q5.28): $[Co(NH_3)_6]Cl_2 \rightarrow [Co(NH_3)_6]^{2+} + 2Cl^-$: 3 ions **(iii)**.
(Q5.29): unpaired: Cr^{3+} 3, Fe^{2+} (weak field) 4, Zn^{2+} 0: highest $[Fe(H_2O)_6]^{2+}$ **(ii)**. **(Q5.30):** chelate $[Fe(C_2O_4)_3]^{3-}$ **(iii)**.

Pattern A: Q5.6, 5.7, 5.24. Pattern B: Q5.5, 5.23. Pattern C: Q5.8–5.12. Pattern D: Q5.15, 5.19. Pattern E: Q5.16–5.18, 5.20, 5.21, 5.25, 5.31. Pattern F: Q5.28–5.30. Theory and give-reasons: Q5.1–5.4, 5.13, 5.14, 5.22, 5.26, 5.27.

Intext Questions 5.1–5.5

Formulas and names (5.1, 5.2) follow Pattern A; isomerism (5.3, 5.4) Pattern C; VBT of $[Ni(CN)_4]^{2-}$ and $[NiCl_4]^{2-}$ (5.5) Pattern D.

5C Exam tips

Common mistakes that lose marks

- Ligand order in names is alphabetical by ligand name, not by charge; in formulas, the metal comes first.
- Forgetting "-ate" for anionic complexes, or using Latin names (ferrate, cuprate, argentate, plumbate, stannate, aurate).
- en and $C_2O_4^{2-}$ count as *two* donor sites each when finding the coordination number.
- Square planar complexes (dsp^2) are never optically active.
- Strong-field ligand + d^4–d^7 $\Rightarrow$ low spin (fewer unpaired electrons).

Question types: 1-mark: oxidation state, CN, IUPAC name, ambidentate ligand; 2–3 marks: VBT, isomers, CFT colour; 5 marks: VBT + CFT with magnetic moment, isomer drawings.

5D JEE extras

Useful results

- $\Delta_t = \frac{4}{9}\Delta_o$; tetrahedral complexes are almost always high spin.
- CFSE (octahedral) $= (-0.4\, n_{t_{2g}} + 0.6\, n_{e_g})\Delta_o +$ pairing energy.
- EAN $= Z -$ (oxidation state) $+ 2\times$CN; carbonyls follow 18-electron rule.
- Number of geometrical isomers: $[Ma_2b_2]$ (planar) 2; $[Ma_4b_2]$ 2; $[Ma_3b_3]$ 2; $[M(AA)_2b_2]$ cis (optically active) + trans.

Example: $[Fe(CN)_6]^{4-}$: EAN $= 26 - 2 + 12 = 36$ (Kr).

5E Pattern spotter

If you see…	Do this	See
Naming and counting		
Formula given, name asked	Ligands alphabetical, metal (-ate), O.S.	Ex A
Oxidation state	Sum of charges = charge of entity	Ex B
Coordination number	Count donor atoms (en = 2)	Ex B
Ions in solution	Count outside the bracket	Ex F
Structure		
Number of isomers	Planar / octahedral rules; check mirror image	Ex C
Hybridisation, magnetism (VBT)	Strong field pairs; inner vs outer orbital	Ex D
Colour, λ order	Spectrochemical series; $\Delta_o \uparrow \Rightarrow \lambda \downarrow$	Ex E
Most stable complex	Chelating, strong-field ligand	Ex F